

VEGGLI II, Norway

WMO: 014710

Lat: 60.0435N

Lon: 9.1468E

Elev: 902

StdP: 14.22

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.1	(c) 4.3	(d) -5.2	(e) 4.3	(f) 1.1	(g) -1.1	(h) 5.4	(i) 5.8	(j) 15.2	(k) 39.1	(l) 12.9	(m) 37.9	(n) 3.7	(o) 280	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	80.1	60.4	75.9	59.1	72.0	57.5	63.0	73.1	61.5	70.9	59.8	68.4	5.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	79.4	65.0	58.0	74.2	63.8	56.2	69.7	62.3	28.9	73.6	27.8	71.6	26.6	68.1	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.3	8.8	-3.4	85.4	5.2	3.3	-7.1	87.8	-10.1	89.7	-13.0	91.6	-16.8	94.0
			-3.9	66.1	5.1	2.3	-7.6	67.7	-10.6	69.1	-13.4	70.4	-17.1	72.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	23.2	25.5	33.2	41.2	49.7	56.7	60.9	57.4	51.1	40.6	31.3	25.4		
	DBStd	14.91	9.47	9.47	7.75	6.32	7.32	5.41	5.27	4.01	4.94	6.09	6.87	10.32		
	HDD50	4086	831	686	522	273	98	6	0	2	46	297	562	763		
	HDD65	8638	1296	1106	987	715	476	255	152	238	418	755	1012	1228		
	CDD50	959	0	0	1	8	88	208	339	230	78	7	0	0		
	CDD65	37	0	0	0	0	2	7	27	1	0	0	0	0		
	CDH74	530	0	0	0	0	52	145	318	13	2	0	0	0		
	CDH80	94	0	0	0	0	4	24	66	0	0	0	0	0		
(14) Wind	WSAvg	3.7	3.7	3.7	4.4	4.4	4.2	4.0	3.5	3.1	3.5	3.3	3.3	3.7		
(15) Precipitation	PrecAvg	31.0	2.1	1.4	1.3	1.5	2.5	3.3	3.9	4.1	3.3	3.0	2.8	2.0		
	PrecMax	44.3	4.7	5.1	3.4	3.9	9.2	7.2	9.2	8.0	8.3	8.9	10.9	4.1		
	PrecMin	22.8	0.4	0.2	0.0	0.2	0.4	0.2	0.7	1.0	1.0	0.4	0.3	0.5		
	PrecStd	5.2	1.1	1.0	0.9	1.1	1.9	1.9	2.0	1.7	1.7	1.9	2.2	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.6	50.2	58.2	67.0	79.4	83.3	85.4	75.8	71.6	61.5	50.7	50.4		
		MCWB	41.1	41.9	46.0	50.3	58.2	60.1	62.2	60.5	57.4	54.1	44.5	43.7		
	2%	DB	42.9	46.1	52.7	62.8	74.0	78.4	81.6	72.6	67.3	56.7	47.2	46.3		
		MCWB	37.4	38.6	43.1	48.8	56.1	58.3	61.8	59.7	55.8	49.1	42.3	40.1		
	5%	DB	38.4	42.2	48.4	57.7	69.1	73.6	77.7	69.8	63.7	53.4	43.8	42.4		
		MCWB	33.9	36.2	40.0	45.6	53.7	56.3	61.1	57.9	54.1	47.4	40.4	37.7		
	10%	DB	35.4	38.4	44.6	53.1	64.4	69.3	73.2	66.6	60.4	50.2	40.6	38.3		
		MCWB	32.0	34.0	37.0	43.5	51.8	54.6	59.5	56.3	52.4	45.0	37.5	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	43.3	46.7	52.1	60.7	62.3	65.6	63.7	60.2	55.3	46.4	43.9		
		MCDB	46.8	50.2	57.1	63.4	72.4	73.1	76.5	70.8	66.6	61.0	48.4	50.0		
	2%	WB	37.7	39.4	43.4	49.4	58.2	60.3	63.9	61.9	57.7	51.2	43.3	40.6		
		MCDB	41.9	44.5	52.1	60.3	70.1	72.7	75.3	68.6	63.6	54.6	46.1	45.1		
	5%	WB	34.9	36.7	40.8	46.8	55.6	58.3	62.4	60.1	55.7	48.4	40.8	38.1		
		MCDB	37.6	41.4	48.0	56.1	65.7	69.5	73.3	66.2	61.3	51.9	43.6	41.7		
	10%	WB	32.6	34.4	37.7	44.4	53.0	56.5	60.9	58.2	54.0	46.0	38.3	35.1		
		MCDB	34.8	37.7	43.1	52.5	61.9	66.2	70.5	64.1	58.8	49.6	40.3	37.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	12.2	16.2	18.4	20.2	19.8	18.8	17.8	16.8	13.0	9.6	10.4		
		MCDBR	14.5	16.9	20.6	26.2	27.5	28.0	26.4	22.5	22.2	17.9	14.5	15.5		
	5% WB	MCWBR	10.9	11.4	12.6	14.0	13.1	12.6	10.9	11.2	12.6	11.3	11.3	11.4		
		MCWBR	13.1	14.9	20.0	23.4	23.4	22.3	21.7	18.3	18.9	15.1	13.1	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.274	0.306	0.346	0.349	0.341	0.360	0.352	0.320	0.304	0.253	0.205		
		taud	2.195	2.283	2.278	2.294	2.426	2.477	2.439	2.452	2.530	2.490	2.282	2.091		
	Ebn at Noon		181	235	262	270	279	283	273	267	259	225	171	141		
		Edh at Noon	12	21	29	34	32	31	32	29	23	17	11	9		
(44) All-Sky Solar Radiation	RadAvg		103	315	794	1293	1525	1697	1501	1186	779	365	132	65		
	RadStd		20	69	66	179	179	151	193	97	106	69	32	6		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)		N/A	N/A	+2.63	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page